Q1 - Speed of sound and compression: Recall that the speed of sound for anything is:

$$\frac{dP}{d\rho} = c_s^2 \tag{1}$$

where dP is the change in pressure, $d\rho$ is the change in density, and c_s is the speed of sound. First, show that

$$d\rho = \frac{dP}{c_s^2} \tag{2}$$

This tells us that if we apply extra pressure to something (dP) that the density ($d\rho$) changes by that pressure divided by the speed of sound squared. It may seem like solids don't shrink when we push on them, but they do and the speed of sound in that solid tells us how much. The speed of sound is effectively a stiffness - a higher speed of sound means a smaller change in density for a given applied pressure.

The speed of sound in sea water is 1500 m/s. What is the pressure at the bottom of the Marianas Trench, 11 km below sea level? What is the change in density between the surface and the bottom? What is the *fractional* change in density? And is the water more dense or less dense in the ocean depths? You may assume the density of water is 1000 kg/m^3. The true answer is about 5%, so our estimate is very close!

Starting from $\frac{dP}{d\rho} = c_s^2$, we multiply both sides by $d\rho$ to get $d\rho c_s^2 = dP$, then divide by c_s^2 to get $d\rho = \frac{dP}{c_s^2}$.

For the pressure, if we're 11 km deep, that's 11,000 metres. Each cubit metre of water is 1000 kg, so we have 11 million kg of water on top of each square metre. Multiply by $g \sim 10 m/s^2$ to get the pressure at the bottom of the trench of 1.1×10^8 Pascal. Alternatively, you might have remembered that one atmosphere corresponds almost exactly to 10m of water, so if we're at 11,000m, then pressure should be 1100 times atmospheric.

We can use our expression for the change in density to work it out now we know the change in pressure and the sound speed. We have

$$d\rho = \frac{dP}{c_s^2} = \frac{1.1e8}{1500^2} = 49\frac{kg}{m^3} \tag{3}$$

Since the background density of water is 1000 kg/m^3, the fractional change is $49/1000 = 0.049 = 4.9\%$.

Incidentally, the density of air at sea level is about a thousandth that of water. If we increase the pressure by a factor of a thousand, air will be about as dense as water! The pressure inside a SCUBA tank is about 200 atmospheres. If we dropped a SCUBA tank into the Mariannas trench, it would *implode* on its way down.

Q2 - Two strings are tied together. String two has twice the mass per unit length compared to string one. Someone at the end of string one sends a wave down it. What are the amplitudes of

the reflected and transmitted waves, relative to the incoming wave, when the incoming wave hits the junction?

If the ratio of masses is a factor of 2, then the ratio of speeds will be $\sqrt{2}$. The second, more massive string will have a lower speed.

We know the transmitted amplitude T in terms of the incoming amplitude A is

$$\frac{T}{A} = \frac{2v_t}{v_i + v_t} \tag{4}$$

if we let $v_t = 1$ and $v_i = \sqrt{2}$ (since the incoming velocity is *higher*), we get

$$\frac{T}{A} = \frac{2}{1+\sqrt{2}} = 2\frac{\sqrt{2}-1}{2-1} = 2(\sqrt{2}-1) = 0.82... \tag{5}$$

The reflected fraction is

$$\frac{R}{A} = \frac{v_t - v_i}{v_t + v_i} = \frac{1-\sqrt{2}}{1+\sqrt{2}} = -\frac{(\sqrt{2}-1)^2}{2-1} = 2\sqrt{2} - 3 = -0.17... \tag{6}$$

In both cases, we've used the identity $(a-b)(a+b) = a^2 - b^2$ to turn the denominators into rational numbers. As expeted, because the second string is heavier than the first, the reflected wave has negative amplitude.

Q3 - A violin body has a volume of about 1850 cubic cm. Its f-holes are tuned so the Helmholtz resonance is at 330 Hz. Typical dimensions are about 8mm wide and 75mm long. What is the effective length of the resonating air plug? You should find it's comparable to the width of the f-hole. Remember that the resonant frequency of a Helmholtz resonator is

$$\nu = \frac{v}{2\pi}\sqrt{\frac{A}{LV}} \tag{7}$$

We want to use the resonant frequency of a Helmholtz to find L as we know everything else in the problem. We can square both sides of the expression for the frequency to get:

$$\nu^2 = \frac{v^2}{4\pi^2}\frac{A}{LV} \tag{8}$$

Multiply by L and divide by ν^2 to get

$$L = \frac{v^2}{4\pi^2\nu^2}\frac{A}{V} \tag{9}$$

I actually lied in the problem about the frequency - 330 Hz is an E, not a D, but we'll go with it (the D is 4/3 of an A, so 293 Hz is closer). We set $v = 340m/s$ or so (depending on temperature), and $\nu = 330$, $A = 2 \times 0.008 \times 0.075$ where the factor of 2 comes from having 2 f-holes. The volume given is in cubic centimetres, so we have to divide by 100^3 or one million to convert to m^3. That gives us a length of

$$L = \frac{340^2}{4\pi^3 30^2} \frac{2 \times 0.008 \times 0.075}{1850e-6} = 0.017m \tag{10}$$

That is, the effective length of the resonator is 17mm, or twice the width. This is actually quite a resonable answer - if the hole was say 1cm wide, then the air 1cm above and 1cm below the hole moves as part of the resonator.

Q4 - As mentioned in class, a cracked window in a moving car often leads to a thumping Helmholtz resonance. Would you exect the frequency of the thumping to depend on the speed of the car?

You would not in general expect the frequency to depend on the speed at which you're driving. The frequency is set by the geometry of the resonator. The resonance will get *louder* if we go faster, but the frequency will stay the same. Just like if we blow across a bottle harder, the sound will get louder but the pitch stays the same.

Q5 - You see a flash of lightning. Exactly 10 seconds later, you hear thunder. How far away was the lightning if the day was very hot (+35 C)? Very cold (-10 C)?

The speed of sound in air at 0C/273K is 330 m/s. It scales almost perfectly with the square root of temperature, so the speed of sound at +35C will be

$$330 \times \sqrt{\frac{273+35}{273}} = 350.5m/s \tag{11}$$

On the cold day, we have

$$330 \times \sqrt{\frac{273-10}{273}} = 323.9m/s \tag{12}$$

Since in both cases the thunder takes 10 seconds to get to us, the distances are $350.5m/s \times 10s = 3505m = 3.5km$ on the hot day, and $323.9m/s \times 10s = 3.24km$ on the cold day. The answers agree to better than 10%, so the old rule of thumb that it's about 3 seconds per km is very accurate.